

# Questions for the benchmark

- Does reusing vetted code improve accuracy or reduce the cost of a complete forecasting workflow?
- Our skills teach agents how to use library APIs from Python. weather-skills primarily exposes parameterized commands.
- Working hypothesis: agents need Python to recover from data, parameter, and workflow errors.
- Compare task success, tokens, cost, and completion time across model sizes. Do not assume skills will win.

# Current experiment: three execution conditions

- **Python only:** the agent writes, runs, and revises its own code using standard scientific Python packages.
- **Skills only:** the agent reads guides and invokes weather-skills operations; custom Python is disabled.
- **Python + skills:** the agent can combine catalog operations with its own Python. Skill use is measured, not assumed.
- Ten models, one task, three conditions: 30 planned attempts. The small-model tier includes 1B, 3B, 7B, 8B, and 9B models.

Astra is excluded because it helped develop the benchmark. Earlier runs remain archived.

# Current task: a two-week Kenya heat outlook

- Use a real, archived ECMWF S2S forecast to calculate the hottest daily regional-mean temperature in each week.
- Required order: cosine-latitude spatial mean for each member/day → weekly peak per member → ensemble median, spread, minimum, and maximum.
- Pass requires a structured answer matching an independent numeric reference, correct units and source citation, and a valid, nonblank figure.
- Every attempt gets the same cached raw data and an empty workspace. Intermediate results stay within that attempt; no outputs or history cross runs.

Equal ceilings: 24 model calls, 40 executions, 200k tokens, 600 seconds.

Download and setup are excluded so solve time measures analysis, not data transfer.

# Weather agent benchmark

Heat outlook · 10 models · Astra excluded · Complete

Experiment

Heat outlook · 10 models · Astra excluded · 30/30 · Complete

Model

All models

Task

All experiment tasks

30 recorded attempts · 30 task outcomes · 10 models · 1 tasks · 1 repetition · ≥\$2.713 reported spend · billing incomplete

ResultsTasks 1Runs 30

GPT-6 Astra is excluded from this comparison because it helped develop the benchmark, tasks, and harness. This avoids evaluating the benchmark-authoring model; it does not establish that the remaining evaluation is free of design bias. Earlier Astra runs remain in archived experiments.

30 / 30 attempts recorded

Complete · Updated 12:41 PM

11 passed19 task failures0 provider errors0 unscored0 pending

| Model / task coverage | Kenya heat outlook · frozen real forecast |                     |                         |
|-----------------------|-------------------------------------------|---------------------|-------------------------|
| Claude Fable 5.1      | Python only<br>Pass                       | Skills only<br>Pass | Python + skills<br>Pass |
| Claude Sonnet 5.5     | Python only<br>Pass                       | Skills only<br>Fail | Python + skills<br>Pass |
| Gemini 3.1 Flash-Lite | Python only<br>Pass                       | Skills only<br>Fail | Python + skills<br>Pass |
| DeepSeek V4.1 Flash   | Python only<br>Pass                       | Skills only<br>Pass | Python + skills<br>Pass |
| Qwen3.5 9B            | Python only<br>Pass                       | Skills only<br>Fail | Python + skills<br>Fail |
| Minstral 3.2B         | Python only                               | Skills only         | Python + skills         |

Completed study: 30/30 attempts. Select a result to inspect its trace.

# Observed failure modes

- **Skills bypassed:** 0 of 6 Python + skills attempts invoked a skill. The prompt makes skills optional; this condition tests availability, not actual reuse.
- **Interface friction:** wrong action names, XML tool-call wrappers, guide-read requirements, and invalid CLI arguments consumed turns.
- **Recovery and delivery:** Python indexing and unit errors required retries. Sonnet produced a figure but failed the strict artifact-submission contract.
- **Budget exhaustion:** 6 of 7 failures hit the token limit; Qwen's hybrid run hit the time limit. Repeated context and unusable responses contributed.

Initial six-model batch: 18 attempts, 7 failures. One attempt per model and condition.

## Next: test the libraries and their skill guides directly

- **Baseline:** general scientific Python; the agent implements the workflow itself.
- **Libraries:** the same Python environment plus acmadDL and africaS2S, with ordinary API documentation.
- **Libraries + skills:** the same libraries plus agent guides and progressively disclosed workflow references.
- Keep Python available in every condition. Use seasonal tasks covering data acquisition, downscaling, held-out verification, and forecast delivery.

Predeclare tasks and scoring; freeze versions and seeds; repeat across regions, methods, and models.